

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL

8223/2

PAPER 2

1 hour 15 minutes

Marks 60

2021

Additional Materials: Non-programmable calculator

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- You may use a non-programmable calculator.
- Do not use correction fluid.

- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.
- You will lose marks if you do not show your working or if you do not use appropriate units.

For Examiner's use

1	
2	
3	
4	
5	
6	
Total	

Marker	
Checker	

This document consists of **11** printed pages and **1** blank page.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 Fig. 1.1 shows a cube representing a cell of a certain size.

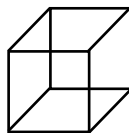


Fig. 1.1

- (a) (i) Use a ruler to measure the side length of the cube in Fig. 1.1.
side length mm [1]
- (ii) Calculate the surface area and the volume of the cube in Fig. 1.1.
surface area mm²
volume mm³ [2]
- (iii) Calculate the surface area to volume ratio of the cube and state your answer to 2 decimal places.
surface area to volume ratio [1]
- (b) State the relationship between cell size and surface area to volume ratio.
.....
..... [1]
- (c) Suggest why the surface area to volume ratio is important to a living cell.
.....
.....
.....
.....
.....
..... [3]
- (d) Name one specialised plant cell and one specialised animal cell that is adapted to have an increased surface area to volume ratio.
Describe the adaptation of each cell that you name.
plant cell
adaptation
.....
animal cell
adaptation
..... [2]

[10]

- 2 Fig. 2.1 shows the molecular structure of a molecule of glucose in the linear form.

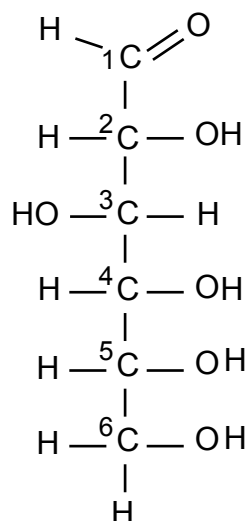


Fig. 2.1

- (a) (i) Draw the molecular structure of a molecule of glucose in the β -ring form.

[3]

- (ii) Describe the difference between the ring structures of α and β -glucose.

.....

.....

[1]

(b) Fig. 2.2 shows the molecular structure of cellulose.

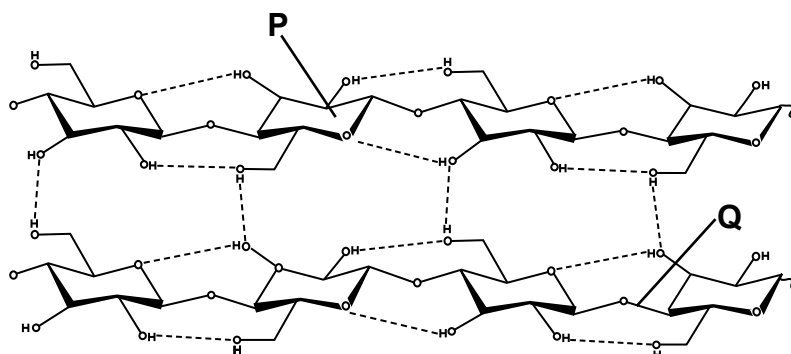


Fig. 2.2

(i) Identify the parts labelled **P** and **Q** in Fig. 2.2.

P

Q [2]

(ii) State the importance of the hydrogen bonds shown in Fig. 2.2 on the structure of cellulose.

.....

..... [1]

(iii) Hydrogen bonds also form between molecules of water.

Outline the formation of hydrogen bonds between molecules of water.

.....

.....

.....

.....

.....

..... [3]

[10]

- 3 Fig. 3.1 shows a photomicrograph of a cell of the immune system.

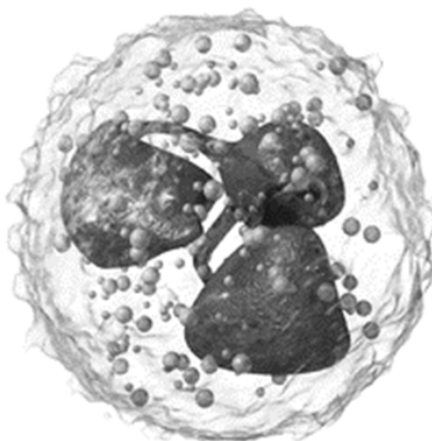


Fig. 3.1

- (a) (i) Identify the cell and describe the structure of the type of cell shown in Fig. 3.1.

.....

.....

.....

.....

.....

.....

[3]

- (ii) Describe the role in immunity of the type of cell shown in Fig. 3.1.

.....

.....

.....

.....

[2]

- (b) Fig. 3.2 shows the activation of B-lymphocytes in an immune response following infection by a pathogen.

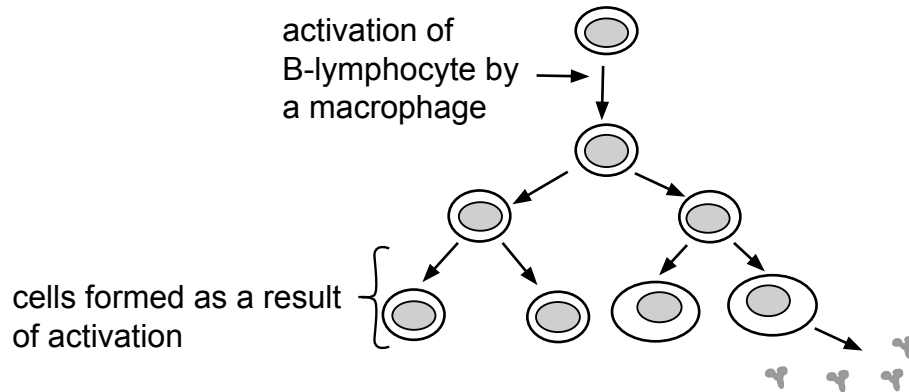


Fig. 3.2

- (i) Outline the role of macrophages in the activation of B-lymphocytes.

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.....

.....

[2]

- (ii) Name and outline the importance of each type of cell formed as a result of the activation of B-lymphocytes.

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[5]

[12]

- 4 Fig. 4.1 is a photomicrograph of part of the human gas exchange system.

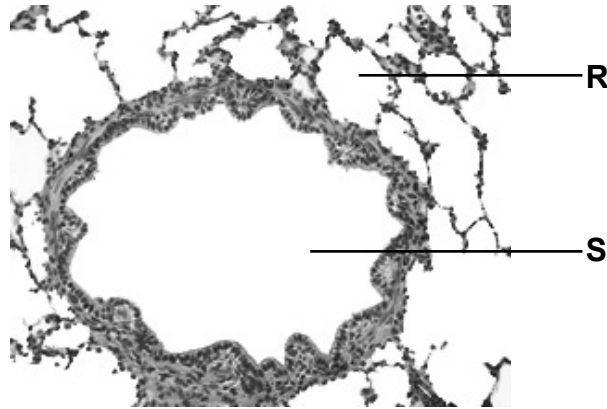


Fig. 4.1

- (a) (i) Identify structures **R** and **S** in Fig. 4.1.

R

S [2]

- (ii) Explain how structure **R** is adapted for gas exchange.

.....

 [3]

- (iii) Describe the function of elastic fibres in the wall of structure **R**.

.....

 [2]

- (b) Inhalation of tobacco smoke can lead to development of the long-term health condition COPD.

State one symptom of COPD.

.....
 [1]

[8]

- 5 Fig. 5.1 shows the structure of a chromosome during metaphase of mitosis.

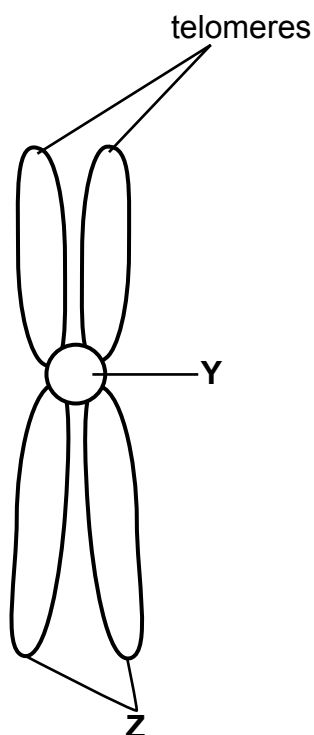


Fig. 5.1

- (a) Identify the parts labelled **Y** and **Z** in Fig. 5.1.

Y.....

Z

[2]

- (b) Describe a telomere and outline the significance of telomeres in DNA replication.

.....

.....

.....

.....

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.....

.....

[4]

- (c) Describe what happens over time to the telomeres of chromosomes in a eukaryotic cell.

.....

.....

[1]

(d) Telomerase is an enzyme that catalyses the formation of telomeres.

Telomerase is present in only a very low concentration in eukaryotic cells but is present at a very high concentration in cancer cells.

Suggest how a very high concentration of telomerase enzyme enables cancerous cells to form a tumour.

.....

.....

.....

.....

.....

.....

[3]

[10]

- 6 Fig. 6.1 illustrates two processes, **one** and **two**, that move substances into the cells of a plant root.

The net movement of a substance during each process is shown by an arrow.

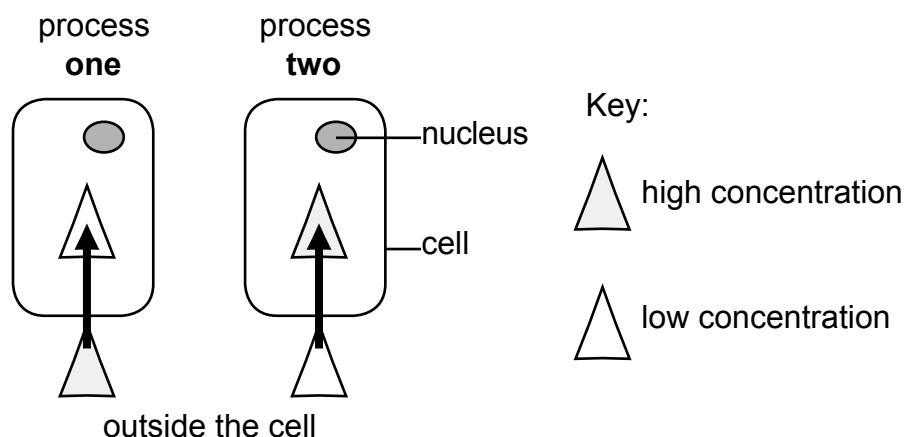


Fig. 6.1

- (a) (i) Complete Table 6.1 to identify which process, **one** or **two**, transports each named substance into the cell and state evidence from Fig. 6.1 to support your answer.

Table 6.1

substance transported	process	supporting evidence from Fig. 6.1
water		
mineral ions		

[2]

- (ii) Hydrogen cyanide is a toxin that prevents respiration from taking place.

Suggest and explain the effect of hydrogen cyanide on the uptake of water and of mineral ions by the cells of a plant root.

.....

.....

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.....

.....

[3]

(b) Fig. 6.2 shows a cross-section of one vascular bundle in the stem of a plant.

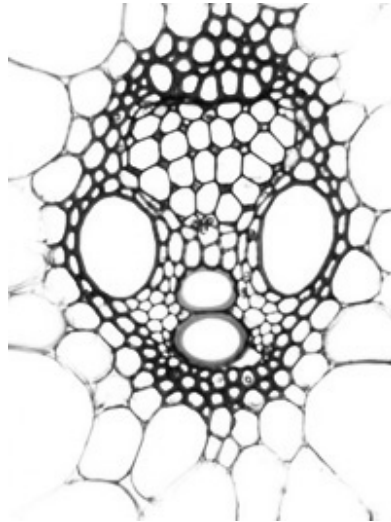


Fig. 6.2

(i) Use a label line to label the tissue that transports assimilates from source to sink, on Fig. 6.2. [1]

(ii) Name the process used to transport assimilates from source to sink. [1]

.....

(iii) Outline how sucrose is loaded into the tissue you have labelled in **(b)(i)**.

.....

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.....

.....

.....

[3]

[10]

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